

This document presents the analysis report conducted to address the research question: “To what extent is there a systematic misalignment between perceived productivity and actual indicators of rest and well-being?” In a world where productivity - often toxic—is frequently used as an indicator of success and a life “on track,” the author examines the degree of misalignment between perceived productivity and actual rest and well-being indicators. An initial analysis is conducted on the Student Mental Health dataset provided by this assessment. This dataset does not specifically contain data on productivity or actual rest indicators, but the data on GPA (performance, productivity) and mental health status allow us to conduct an indirect analysis of the relationship between apparent productivity and well-being. The analysis specifically addresses the question of to what extent academic performance can serve as a reliable indicator of well-being, and whether there is a discrepancy between academic performance and mental health indicators.

1. Methodology

The assumptions used in this analysis are as follows:

H1: There is no positive relationship between academic performance and improved mental health outcomes among students.

H2: A significant proportion of high-achieving students exhibit symptoms of poor mental health (e.g., anxiety, depression, or panic attacks).

H3: There is no significant difference in mental health outcomes between high-achieving and lower-achieving students.

The dataset used contains 101 entries and 11 columns providing information on students’ mental health, their majors, and their GPA ranges.

In addition to the variables already present in the dataset, three variables were created to measure misalignment. Operationally, “misalignment” here refers to students who perform well academically but show signs of mental distress.

These three variables are:

- Performance group categories: GPA in this dataset is reported as categorical CGPA ranges. We created groups based on the ranges.
- Mental health risk: this column flags profiles that report at least one mental health issue
- Misalignment: this represents students who perform well academically but show signs of mental distress.

A descriptive and comparative analysis was developed to test the hypotheses formulated at the start of the project. Here are the results.

2. Results

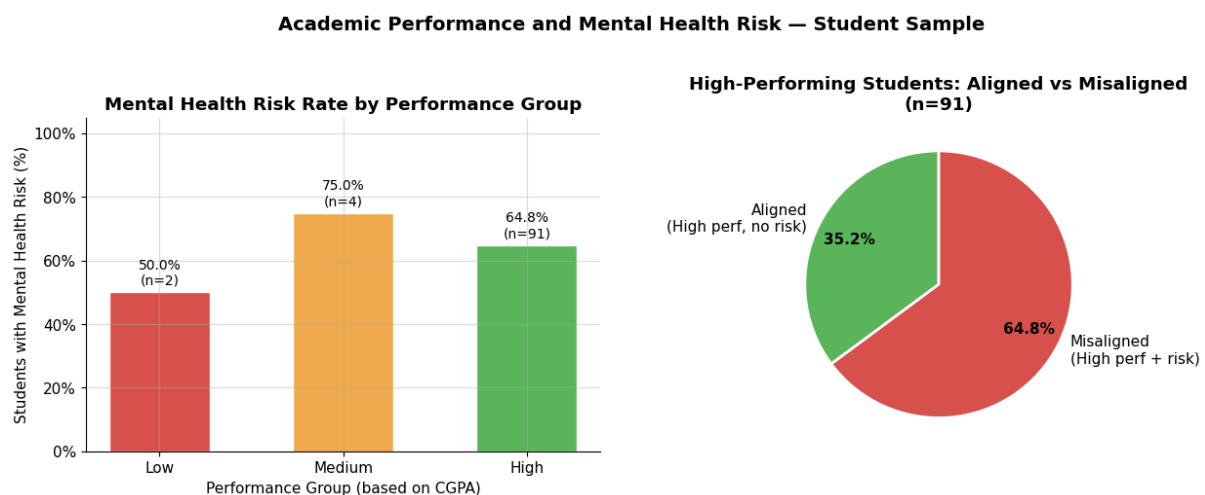
2.1. Descriptive findings

High-performing students account for 91.8% of the entire dataset. Average-performing students account for 4.1%, and low-performing students account for only 2.1%.

We find that students with at least one mental health issue account for 64.9% of the dataset. The proportion of these mental health issues is nearly equal across the dataset (depression 36.1%, anxiety 35.1%, panic attacks 33.0%).

2.2. Core analysis

First, we assessed the mental health risk among each performance group. The analysis shows that students suffer from mental health problems regardless of their academic performance levels. In particular, 64.8% of high performers were found to be at risk of having mental health problems, whereas 75% of average performers and 50% of low performers had mental health problems.



The small sample size of the group of students with low academic performance ($n = 2$) affects the validity of the comparison made above. However, this clearly confirms that high academic performance does not reduce the risk of suffering from mental health disorders.

This analysis supports Hypothesis H1, which states that academic performance cannot serve as an indicator of mental health, as high academic achievement is not associated with good mental health. Furthermore, the data support

Hypothesis H3, indicating that there are no significant differences in mental health among the different groups of students based on their performance levels.

Overall, the dataset shows a misalignment rate of 60.8%, indicating that for more than half of the students, academic performance is not an indicator of mental well-being.

This high misalignment rate supports Hypothesis H2: a significant number of high-performing students suffer from psychological disorders, meaning that academic success does not serve as an indicator of well-being.

To test hypotheses H1 and H3 in greater detail, we conducted a statistical analysis of these hypotheses. The p-value obtained with the chi-square test is significantly greater than α ($\alpha = 0.05$ and p-value = 0.8293), leading us to conclude that academic performance cannot be used to predict mental health risk.

The chi-square test was selected for this correlation analysis due to the two variables concerned - performance group and mental risk group - being categorical.

Furthermore, a comparison of mental health risks between high-performing students ($\text{GPA} \geq 3.00$) and low-performing students ($\text{GPA} < 2.50$) showed that high-performing students had a higher rate of mental health risks than low-performing students (a difference of 14.8 percentage points). This implies an imbalance between academic performance and mental health among students. Given that hypothesis H1 states that there is NO relationship between high academic performance and good mental health, we can confirm that this claim is supported by the data.

2.3. Discussions

These findings highlight a significant discrepancy between academic achievement and mental health in the students. In general, 64.9% of the students present at least one symptom of psychological disorders, such as depression, anxiety, and panic attacks. This shows that psychological distress is globally prevalent among these students, irrespective of their academic performance.

In accordance with Hypothesis 1, there is no correlation between high academic performance and positive indicators of mental well-being. Indeed, the results show that students with a high academic performance have a mental health risk rate of 64.8% while their counterparts have a risk rate of 50.0%. Academic performance, which can be assimilated to productivity, instead of having a

positive effect on psychological well-being, might serve as an additional stressor rather than an advantageous characteristic in the students' lives.

In accordance with Hypothesis 2, a considerable number of students with high academic performance exhibit symptoms of poor mental health. As was mentioned earlier, almost two-thirds (64.8%) of these students have experienced at least one symptom of a psychological disorder. These findings suggest that the existence of the mentioned misalignment phenomenon: students who meet or exceed academic expectations are simultaneously experiencing meaningful levels of psychological distress.

Hypothesis 3 posited that there is no significant difference in mental health outcomes between high-performing and lower-performing students. The result from the chi-square test confirmed the hypothesis, meaning that the performance alone is not a factor affecting mental health risk.

From all the above we can conclude that academic performance does not determine well-being. There is a considerable percentage of students who have high CGPA but poor mental health. Within the scope of our broader research question, perceived productivity—often regarded as an indicator of a satisfying life—does not allow us to assess true indicators of well-being or actual needs.

3. Limitations

We acknowledge that this dataset does not specifically provide information on perceived productivity or an assessment of well-being indicators, as intended in the overall research question. We conducted a preliminary analysis using indirect variables whose statistical potential remains undeniable. Furthermore, given its size and the uneven distribution of academic performance, the dataset may not be sufficiently representative. Finally, bias could have been introduced into the data, since reported mental health issues may have been self-assessed rather than based on a proper diagnosis.

In conclusion, academic achievement is not an effective tool for measuring the welfare of students. With almost two-thirds of the high-achievers reporting mental health issues, and no statistical significance in the psychological outcome among the performance groups, the analysis reveals that high CGPA and poor mental health coexist. On a wider perspective in the study, the result supports the notion that the rest needs, and well-being cannot be assessed by their productivity levels.